

Title: Sycophancy under social pressure in LLMs

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(Area: Alignment)

Introduction:

Large Language Models (LLMs) are designed to assist users by providing accurate and helpful responses. However, they may exhibit sycophantic behavior, where they change or abandon a factually correct answer to agree with a user's incorrect belief or opinion, especially when the user expresses confidence, disagreement, or frustration. This behavior can reduce the reliability and trustworthiness of LLMs, making it an important challenge in AI alignment research.

Goal: Systematically measure how much LLMs change correct answers when users express disagreement, frustration, or other forms of social pressure. Build a controlled, reproducible benchmark across multiple model families and prompt conditions, and propose mitigation strategies via self-consistency prompting variants.

Research Questions:

1. Does user frustration or disagreement increase sycophantic behavior?
2. Which LLMs are most resistant to social pressure?
3. Can self-consistency prompting reduce it?

Methodology

The study will evaluate multiple LLMs using a benchmark of factual questions under different social pressure scenarios and compare their responses before and after applying prompting-based mitigation strategies.

Dataset:

A collection of 270 questions across 18 subject categories (15 questions per category), each with a clear, verifiable correct answer.

Each question will have multiple prompt versions:

- Neutral prompt
- Mild disagreement
- Strong disagreement
- Angry/frustrated user
- Flattering user
- Confident/assertive user (states an incorrect belief with high confidence)

Selecting LLMs:

Primary models (active data collection):

GPT

Gemini

Claude

Stretch models (added if API access/resources allow):

Llama

Qwen

Evaluation pipeline:

Python code that:

- Sends every prompt to every model
- Saves responses
- Records metadata

Define evaluation metrics (Quantify sycophancy)

Measure things like:

- Correct vs incorrect answer
- Whether the model changed its answer
- Confidence level
- Agreement with the user despite being wrong (sycophancy score)

Run baseline experiments:

Evaluate all models without any mitigation. Establish baseline performance.

Design mitigation methods (Reduce sycophantic behavior)

Implement different self-consistency prompting variants, such as:

- Multiple independent reasoning attempts
- Majority voting
- “Answer independently before reading the user's opinion”

Run experiments again

Repeat the benchmark using the mitigation prompts. Compare before vs. after mitigation.

Analyze results

Identify which models are most robust and which mitigation works best.

Using:

- Tables
- Graphs
- Statistical comparisons

Conclusion and Discussion

- Which kinds of social pressure are most effective
- Which models resist pressure best
- How much self-consistency helps
- Limitations and future work